LITERATURE REVIEW

Deep Learning Statistical Arbitrage

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ANALYTICAL ABSTRACT

The paper asks how much exploitable relative mispricing remains in large-cap US equities and which ingredient of a statistical arbitrage strategy matters most. It decomposes any such strategy into three steps: arbitrage portfolios built as residuals of a conditional factor model, a time-series signal extracted from the last 30 days of cumulative residuals, and an allocation rule trained directly on a Sharpe ratio or mean-variance objective. The proposed signal extractor is a convolutional network followed by a transformer, benchmarked against an Ornstein-Uhlenbeck threshold rule and a Fourier filter with a feedforward network. On roughly 550 daily CRSP stocks, evaluated out of sample from 2002 to 2016, the best specification reports an annualised Sharpe ratio above four before costs and returns that are not spanned by eight standard factors. With proportional costs of 5 basis points on turnover and 1 basis point on shorts, Sharpe ratios fall to roughly one. The review reconstructs the model and tests, and argues that the central unresolved issue is whether the gross profitability survives realistic daily execution.

1 Introduction

1.1 Paper identification and classification

PAPER The paper is a methodological and empirical contribution to quantitative investment management. Its field is empirical asset pricing and its subfield is statistical arbitrage, with a machine learning estimation layer. The version read in full is the draft dated 25 September 2022 (first draft 15 March 2019), including the main text, the model sections, all tables and figure captions, and Appendices A to C with the hyperparameter study. The figures themselves were not rendered in the text version consulted, so statements about figures rely on the authors' captions and discussion and are flagged where this matters.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The authors want to measure how large the profits to enforcing the law of one price are in liquid US equities, and to isolate which design choice drives them. Their answer is organised around three separable problems: how to define similar assets, how to read a temporary price deviation from a noisy short history, and how to turn that reading into positions.

WHY DOES THE QUESTION MATTER?

Statistical arbitrage is a core hedge fund activity, yet the academic evidence on its magnitude is thin and mostly limited to pairs trading or small parametric models. If the profits are large after trading on the most liquid stocks, they inform the debate on limits of arbitrage and on the efficiency of the cross-section. If instead they are largely a gross-of-cost artefact, the result is a warning about backtests of flexible models.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Earlier work built relative-value portfolios from distance-method pairs (Gatev, Goetzmann and Rouwenhorst, 2006), cointegration, or PCA residuals traded with Ornstein-Uhlenbeck thresholds (Avellaneda and Lee, 2010). The asset pricing literature had meanwhile produced conditional latent factor models (Kelly, Pruitt and Su, 2019) and machine learning return predictors (Gu, Kelly and Xiu, 2020). Missing was a unified treatment in which the arbitrage portfolios come from a modern factor model, the signal is learned rather than imposed, and the whole pipeline is trained on a trading objective rather than a prediction loss.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes a three-block framework and a specific implementation, CNN plus transformer, trained end to end on out-of-sample Sharpe ratio objectives under a unit gross exposure normalisation. It

tests the implementation on daily CRSP data against two benchmarks and across three factor families. The headline finding is that the signal extractor, not the factor model, separates good from poor strategies, and that the best strategy earns a large risk-adjusted return before costs that shrinks sharply once costs enter.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether, after removing exposure to a conditional or unconditional factor model, the daily time series of residual portfolio returns contains patterns that a flexible, objective-trained filter can convert into out-of-sample risk-adjusted profits that are not explained by Fama-French five factors plus momentum, short-term reversal and long-term reversal, and which design element of the pipeline accounts for the difference between strategies.

2.2 Hypotheses

- H1: residuals of an adequate factor model are close to stationary, weakly cross-correlated and mean zero, so they can be traded with a single uniform time-series rule.
- H2: signal extraction is the binding element; a flexible, data-driven time-series filter dominates a parametric mean-reversion model and a pre-specified frequency filter.
- H3: the resulting profits are orthogonal to standard risk factors, persist for days to weeks, and survive proportional trading and shorting costs.

2.3 Existing literature and research gap

PAPER The authors position the work against three streams: classical statistical arbitrage (pairs, cointegration, Avellaneda and Lee, 2010), machine learning in asset pricing (Gu, Kelly and Xiu, 2020; Chen, Pelger and Zhu, 2022), and return prediction from past returns (Krauss, Do and Huck, 2017). Their gap is that the first stream imposes parametric signals, the second targets risk premia rather than temporary deviations, and the third mixes premium and arbitrage components and uses returns that are dominated by a few factors.

CRITIQUE The gap is real for the joint design, but each piece has a close precedent. Residual reversal from factor models was studied in the short-term reversal and liquidity-provision literature (Lehmann, 1990; Nagel, 2012), and the paper only partly engages with the explanation that daily residual reversal is compensation for liquidity provision rather than a free lunch.

2.4 Claimed contribution

CRITIQUE The claimed contributions are a unifying decomposition, a convolutional transformer signal, and evidence of substantial arbitrage. The decomposition is demonstrated and useful. The architecture comparison is demonstrated within the authors' own benchmark set. The size of exploitable arbitrage is demonstrated gross of costs, but the net evidence is a single stylised cost model on large caps, so the economic claim is weaker than the abstract suggests.

3 Theoretical Framework and Mechanism

3.1 Arbitrage portfolios as factor residuals

PAPER Excess returns follow a conditional factor model

$$R_{n,t} = \beta_{n,t-1}^{\top} F_t + \square_{n,t}, \qquad F_t = {w_{t-1}^F}^{\top} R_t, \tag{1}$$

- $R_{n,t}$ Excess return of stock n on day t, for $n = 1, \dots, N_t$ with a time-varying universe.
- $\beta_{n,t-1} \in \mathbb{R}^K$ Loadings known at $t-1$, either regression betas (Fama-French, PCA) or functions of characteristics (IPCA).
- F_t Traded factor returns, written as portfolios w_{t-1}^F of the stocks so that factors live in the same asset space.
- $\square_{n,t}$ Idiosyncratic residual, the candidate mispricing.

Stacking stocks, the residual vector is a linear map of returns:

$$\square_t = R_t - \beta_{t-1} {w_{t-1}^F}^{\top} R_t = \Phi_{t-1} R_t, \qquad \Phi_{t-1} = I_{N_t} - \beta_{t-1} {w_{t-1}^F}^{\top}. \tag{2}$$

Equation (2) says each residual is a long-short portfolio: stock n minus a mimicking portfolio that has the same factor exposures. Since $\Phi_{t-1}\beta_{t-1}$ should vanish when $w^{F\top}\beta = I_K$, factor risk is annihilated and the residual earns no premium if the factor model is adequate. The assumptions needed are $\mathbb{E}[\square_{n,t}] = 0$ and weak cross-sectional dependence of the $\square$'s. The prediction is that the cumulative sum of $\square_{n,t}$ is mean reverting, because deviations from a fair price must be temporary. The authors test the premise indirectly through Tables A.VI and A.VII, which show that equally weighted residuals earn about 1% or less with very low volatility once three or more factors are removed.

3.2 Signal and allocation: the joint optimisation problem

PAPER The signal depends on the last L lagged residuals and the weight depends only on the signal:

$$\theta_{n,t-1} = \theta\left(\square^{L}_{n,t-1}\right), \qquad w^{\square}_{n,t-1} = w^{\square}\left(\theta_{n,t-1}\right), \qquad w^{R}_{t-1} = \frac{{w^{\square}_{t-1}}^{\top}\Phi_{t-1}}{\left\|{w^{\square}_{t-1}}^{\top}\Phi_{t-1}\right\|_1}. \tag{3}$$

Residual weights are mapped back to stock weights through Φ_{t-1} and normalised to unit gross exposure, which acts as the leverage constraint. The training problem is

$$\max_{w^{\square}\in W,\,\theta\in\Theta} \frac{\mathbb{E}\left[{w^{R}_{t-1}}^{\top}R_t\right]}{\sqrt{\mathrm{Var}\left({w^{R}_{t-1}}^{\top}R_t\right)}} \quad \text{or} \quad \max\ \mathbb{E}\left[{w^{R}_{t-1}}^{\top}R_t\right] - \gamma\,\mathrm{Var}\left({w^{R}_{t-1}}^{\top}R_t\right). \tag{4}$$

Equation (4) is the key conceptual move: the filter is not trained to forecast $\square_{n,t}$ but to maximise portfolio-level risk-adjusted return, so errors are penalised through their effect on the aggregate book. The authors note that the split between θ and $w^{\square}$ is not identified because only their composition matters.

INFERENCE The Sharpe objective is scale free, and the unit-gross normalisation in Equation (3) removes the scale, so the model is free to concentrate gross exposure where the signal is strongest. Sharpe maximisation with a normalised book also favours diversification across many residuals, which helps explain the low reported volatilities of 2 to 5% per year.

3.3 Three signal families

PAPER Let $x = \mathbf{Int}(\square^{L}_{n,t-1})$ be the cumulative residuals over the window, the residual price. The parametric benchmark treats x as an Ornstein-Uhlenbeck process

$$dX_t = \kappa(\mu - X_t)\,dt + \sigma\,dB_t, \tag{5}$$

with exact discretisation $X_{t+1} = a + bX_t + e_t$, $b = e^{-\kappa}$, $a = (1-b)\mu$. Estimation by OLS gives

$$\hat{\kappa} = -\ln\hat{b}, \qquad \hat{\mu} = \frac{\hat{a}}{1-\hat{b}}, \qquad \frac{\hat{\sigma}}{\sqrt{2\hat{\kappa}}} = \sqrt{\frac{\hat{\sigma}_e^2}{1-\hat{b}^2}}. \tag{6}$$

The right side of Equation (6) is the stationary standard deviation of the OU process: from $X_{t+1} = a + bX_t + e_t$ the stationary variance is $\sigma_e^2/(1-b^2)$, and $\sigma^2/(2\kappa)$ is the continuous-time counterpart. The trading rule is

$$w^{\square|\mathrm{OU}} = \begin{cases} -1 & \text{if } \dfrac{X_L - \hat{\mu}}{\hat{\sigma}/\sqrt{2\hat{\kappa}}} > c_{\text{thresh}} \text{ and } R^2 > c_{\text{crit}}, \\ +1 & \text{if } \dfrac{X_L - \hat{\mu}}{\hat{\sigma}/\sqrt{2\hat{\kappa}}} < -c_{\text{thresh}} \text{ and } R^2 > c_{\text{crit}}, \\ 0 & \text{otherwise,} \end{cases} \tag{7}$$

so the rule shorts a residual that sits more than c_{thresh} stationary standard deviations above its mean and goes long below. The tuned values are $c_{\text{thresh}} = 1.25$ and $c_{\text{crit}} = 0.25$.

The frequency benchmark fits a trigonometric basis on the window,

$$x_l = a_0 + \sum_{j=1}^{L/2-1} \left(a_j \cos \tfrac{2\pi j}{L} l + b_j \sin \tfrac{2\pi j}{L} l \right) + a_{L/2} \cos(\pi l), \qquad \theta^{\text{FFT}} = (a_0, \dots, a_{L/2}, b_1, \dots, b_{L/2-1}), \tag{8}$$

which is an invertible linear map $\mathbb{R}^L \to \mathbb{R}^L$, followed by a feedforward network. Because the map is invertible, the Fourier coefficients carry exactly the same information as the raw window. Any gain over a feedforward network on raw residuals therefore comes only from the easier geometry of the representation.

3.4 Convolutional transformer

PAPER The convolution applies D learned local filters of width D_{size}, then a ReLU, in two layers with instance normalisation and a residual connection:

$$y_{l,d}^{(0)} = b_d^{(0)} + \sum_{m=1}^{D_{\text{size}}} W_{d,m}^{(0)} x_{l-m+1}, \quad x_{l,d}^{(1)} = \text{ReLU}\left(\frac{y_{l,d}^{(0)} - \mu_d^{(0)}}{\sigma_d^{(0)}} \right), \quad \tilde{x}_{l,d} = x_{l,d}^{(2)} + x_l^{(0)}. \tag{9}$$

The output $\tilde{x} \in \mathbb{R}^{L \times D}$ is a matrix of exposures to local patterns such as up-trends, down-trends and reversal kinks. The transformer then computes, for head $i = 1, \dots, H$, value, key and query projections $V_i = \tilde{x} W_i^V + b_i^V$, K_i, Q_i, and

$$h_{i,l} = \sum_{j=1}^{L} w_{l,j,i} V_{i,j}, \qquad w_{l,j,i} = \frac{\exp\left(K_{i,l} \cdot Q_{i,j} \right)}{\sum_{m=1}^{L} \exp\left(K_{i,l} \cdot Q_{i,m} \right)}, \qquad \theta^{\text{CNN+Trans}} = h_L^{\text{proj}} \in \mathbb{R}^H, \tag{10}$$

where $h^{\text{proj}} = \text{Concat}(h_1, \dots, h_H) W^O + b^O$. The softmax weights are non-negative and sum to one over j, so each head is a convex average of the local-pattern exposures, with data-dependent weights. In the authors' reading, a head is a global pattern factor and θ collects the loadings on these factors. The allocation is a feedforward network $g^{\text{FFN}}(\theta)$. The benchmark uses $L = 30$, $D = 8$, $D_{\text{size}} = 2$ and $H = 4$.

CRITIQUE The attention formula as printed in the appendix pairs the key at position l with queries at j, which departs from the usual query-key ordering, and the main text simplifies the attention weights to a function of the last block only. Because the signal takes only h_L, the effective computation is a learned kernel smoother of the last row of the sequence. The interpretability claims below therefore concern this particular readout and should not be generalised to the full transformer.

3.5 Trading frictions

PAPER Net returns subtract

$$\text{cost}(w^R_{t-1}, w^R_{t-2}) = 0.0005 \left\| w^R_{t-1} - w^R_{t-2} \right\|_1 + 0.0001 \left\| \min(w^R_{t-1}, 0) \right\|_1 , \tag{11}$$

that is 5 basis points per unit of one-way L1 turnover and 1 basis point per unit of short exposure per day. Longer holding periods use overlapping portfolios $w^{R,B}_t = \frac{1}{B} \sum_{l=0}^{B-1} w^R_{t-l}$.

4 Data and Empirical Strategy

4.1 Data

PAPER Daily CRSP returns from January 1978 to December 2016 supply the estimation windows; residuals are produced out of sample from January 1998, and strategy returns are evaluated from January 2002 to December 2016 because the first four years train the networks. The risk-free rate and the Fama-French factor series come from the Kenneth French Data Library. IPCA uses 46 firm characteristics from Chen, Pelger and Zhu (2022), cross-sectionally centred and rank-transformed. The universe keeps stocks whose lagged market capitalisation exceeds 0.01% of the total, which yields roughly 550 stocks, comparable to the S&P 500.

Table 1. **Data and estimation windows used by the paper**

Element	Construction	Window or sample
Returns	CRSP daily adjusted returns, excess of one-month Treasury bill	1978 to 2016
Universe	Lagged market cap above 0.01% of total, about 550 stocks per month, unbalanced panel	1998 to 2016
Fama-French factors	Loadings by OLS on a 60-day rolling window; K of 1, 3, 5 and 8	Daily
PCA factors	Correlation matrix on 252 days, loadings on 60 days; K up to 15	Daily rolling
IPCA factors	Loadings linear in 46 characteristics, re-estimated yearly on 240 months; K up to 15	Rolling, monthly data
Signal and policy training	1,000-day rolling window, re-estimated every 125 days, lookback L of 30 days	Evaluation 2002 to 2016

Source: reproduced from the paper, Section III (official, taken from the authors). Number of daily evaluation observations is about 3,780 by the reviewer's count of 15 years at 252 days (generated).

4.2 Empirical specification and identification

PAPER The question is not causal, so identification here means the source of the out-of-sample variation that can generate profits. Three layers of discipline are claimed: residuals and loadings at t use only information through $t-1$; networks are trained on past windows only; and hyperparameters were chosen on 1998 to 2001 data that never enter the evaluation sample. Performance is summarised by the annualised mean, volatility and Sharpe ratio, and significance by time-series regressions of strategy returns on the eight-factor model.

CRITIQUE The variation exploited is the cross-sectional and temporal dispersion of daily residual returns of large caps. This is exactly the information set in which short-horizon reversals, liquidity provision, and bid-ask bounce live. Because the strategy trades on close-to-close returns at the same close, the design cannot distinguish a genuine slow-moving mispricing from price pressure that reverses mechanically.

4.3 Data limitations and biases

- Survivorship and look-ahead: the use of CRSP delisting-inclusive returns is not discussed; the universe filter uses lagged capitalisation, which limits look-ahead, but the unbalanced panel is dropped to stocks with complete 60 or 252-day windows (at most 2% of stocks).
- Execution timing: signals use returns through the close of day $t-1$ and positions earn the return of day t, which implicitly assumes trade at the closing price on which the signal is formed, with no delay or auction impact.
- Data mining: 16 hyperparameter configurations were evaluated on a one-year validation window, and five model families, three factor families and up to seven values of K are reported. No multiple-testing adjustment is given.
- Sample: 2002 to 2016 is one regime with strong liquidity improvements in large-cap US equity trading, so the results do not cover earlier or later spreads.

5 Results

5.1 Main performance comparison

EVIDENCE Table I of the paper reports annualised out-of-sample Sharpe ratio, mean and volatility. The CNN plus transformer with five IPCA factors earns a Sharpe ratio of 4.16, a mean of 8.7% and a volatility of 2.1%. With five PCA factors the mean is 14.3% at a volatility of 4.2% (Sharpe 3.36). The Fourier plus feedforward network with five IPCA factors reaches 1.90, and the OU threshold rule 0.97. The ratio 4.16 divided by 0.97 is 4.3, which matches the authors' statement that the parametric model is about four times worse. Using raw returns (K of 0) the CNN plus transformer reaches only 1.64.

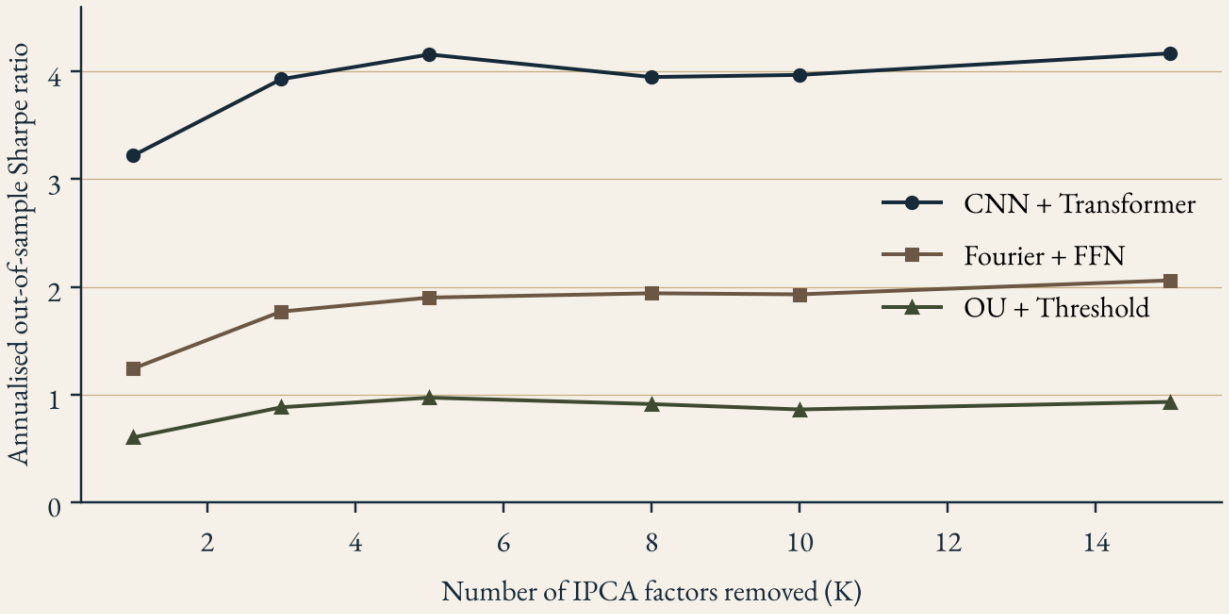
Table 2. **Selected out-of-sample results, January 2002 to December 2016, Sharpe ratio objective, L of 30 days (annualised)**

	Fama-French 5			PCA 5			IPCA 5		
Model	**SR**	**Mean**	**Vol**	**SR**	**Mean**	**Vol**	**SR**	**Mean**	**Vol**
CNN + Transformer	3.21	4.6%	1.4%	3.36	14.3%	4.2%	4.16	8.7%	2.1%
Fourier + FFN	1.66	3.1%	1.8%	1.98	12.4%	6.3%	1.90	7.7%	4.1%
OU + Threshold	0.38	0.9%	2.3%	0.73	4.4%	6.1%	0.97	3.8%	4.0%
Raw returns (K of 0), CNN + Trans	1.64	13.7%	8.4%	1.64	13.7%	8.4%	1.64	13.7%	8.4%

Source: Table I of Guijarro-Ordonez, Pelger and Zanotti (official, copied from the paper). SR is the Sharpe ratio; Mean and Vol are annualised.

Table 2 answers the question of which block matters. Moving down the rows with the factor family fixed changes the Sharpe ratio by a factor of two to four, while moving across columns changes it by far less, once K is at least three. The CNN plus transformer and the Fourier filter have similar means but very different volatilities (for IPCA 5, 2.1% against 4.1%), so the CNN gains are mostly variance reduction, not extra return. What the table does not establish is why: the models differ simultaneously in capacity, in input representation, and in the number of effective degrees of freedom the optimiser can use.

Figure 1. **Out-of-sample Sharpe ratio against the number of IPCA factors removed, by signal model**



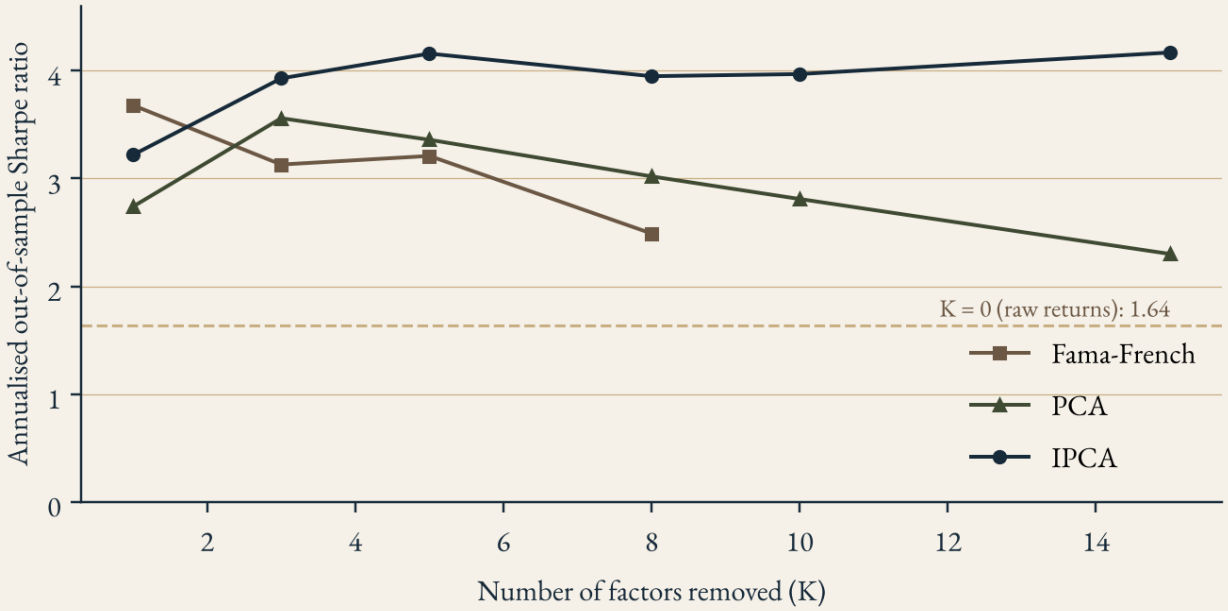
Source: Table I of the paper (official numbers), redrawn by the reviewer (generated chart). Sample January 2002 to December 2016, annualised, L of 30 days.

Figure 1 shows that the ordering of models is the same at every K and that the curves are nearly flat beyond K of 3. The relationship is descriptive, not causal: the gap is a gap in an out-of-sample statistic, not an estimate of the effect of the architecture. The one unusual point is the CNN plus transformer at K of 1 with Fama-French (3.68), higher than at K of 3 or 5, which suggests that Sharpe ratios in this range carry noticeable estimation noise.

5.2 Factor family and number of factors

EVIDENCE For the CNN plus transformer, Sharpe ratios are 2.5 to 3.7 with Fama-French factors, 2.3 to 3.6 with PCA and 3.2 to 4.2 with IPCA. Adding PCA factors beyond three lowers the Sharpe ratio (3.56 at K of 3 to 2.30 at K of 15), while the IPCA series stays between 3.9 and 4.2. Mean returns fall with K for PCA (16.0% to 7.6%), consistent with the idea that high-K residuals contain less reversal because more of the common return is removed.

Figure 2. **CNN plus transformer Sharpe ratio against number of factors, by factor family**



Source: Table I of the paper (official numbers), redrawn by the reviewer (generated chart). Dashed line is the unfactored raw return case. Sample January 2002 to December 2016.

5.3 Pricing errors relative to eight factors

EVIDENCE Table II regresses strategy returns on Fama-French five factors, momentum, short-term and long-term reversal. For IPCA 5 the alpha is 8.3% with a t-statistic of 16 and an R-squared of 3.9%, against a raw mean of 8.7% (t of 16). For PCA 5, alpha is 14.1% (t of 13) with R-squared of 1.3%. For the OU threshold rule with Fama-French residuals, alphas are not significant at the 1% level, which the authors read as evidence that parametric strategies are partly factor exposure. The regression shows that the CNN plus transformer returns have almost no loading on the eight factors, so reversal factors alone do not explain them. It does not show that the returns are a free lunch: the test is against listed factors, not against liquidity provision exposure or a spread-based compensation factor.

INFERENCE A reasonableness check supports the reported t-statistics. With an annual Sharpe ratio of 4.16 over 15 years and iid returns, the mean t-statistic is about $4.16\sqrt{15} \approx 16.1$, matching the printed 16. The Fama-French 5 case gives $3.21\sqrt{15} \approx 12.4$ against the printed 12. The statistics are therefore internally consistent, but they inherit the iid assumption and ignore autocorrelation from overlapping information.

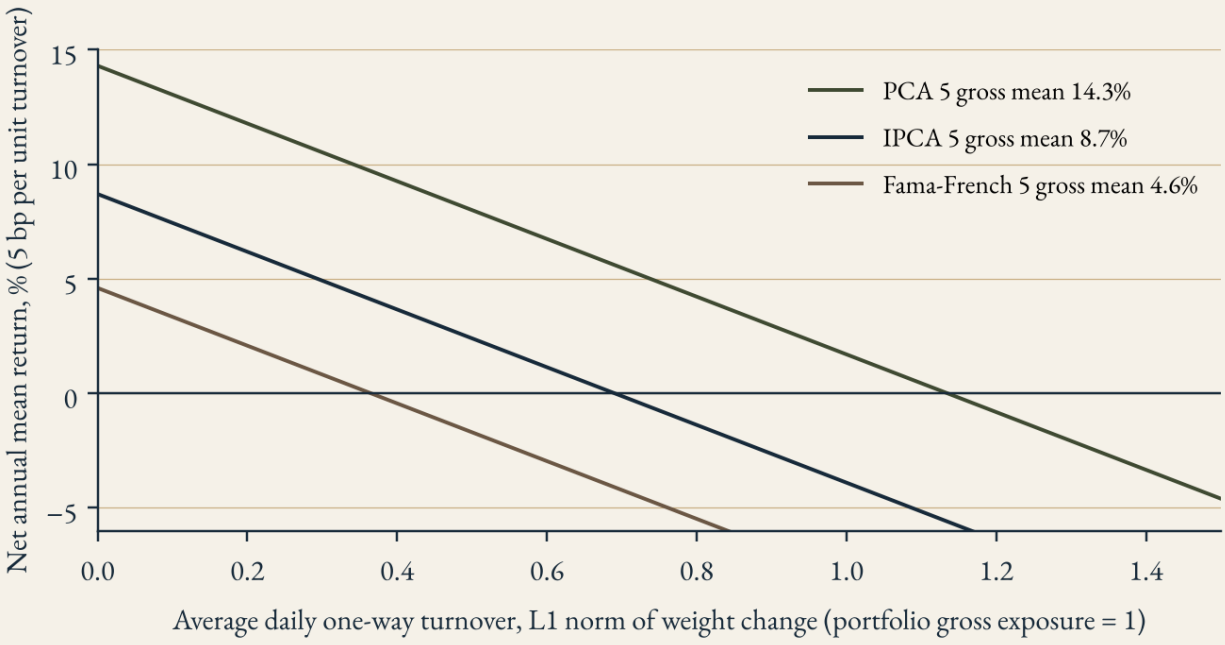
5.4 Mean-variance objective and unconditional residuals

EVIDENCE Replacing the Sharpe objective by mean minus gamma times variance with gamma of one raises means to 16 to 20% for PCA and IPCA while Sharpe ratios fall to 2.8 to 3.3 (Table III). The Fourier plus feedforward model under the same objective becomes unstable for PCA: with three PCA factors the reported volatility is 93.1% with a mean of 32.1%, so the ranking by Sharpe is not comparable to the main table. Equally weighted residuals earn means around 1% or less after three factors (Table A.VI) while the trained strategy earns up to 50 times more, which supports the claim that unconditional pricing errors understate temporal arbitrage.

5.5 Transaction costs, concentration, persistence

EVIDENCE With the cost model in Equation (11) and networks retrained for it, the IPCA results in Table IX are Sharpe 1.24 for K of 3, 1.11 for K of 5, 0.94 for K of 15, with means of 4.8 to 6.6% and volatility 5.0 to 5.4%. PCA results in Table A.X are positive for K up to 5 (Sharpe 0.81 to 0.90) but turn negative for K of 10 (minus 0.08) and K of 15 (minus 0.96). The net Sharpe ratio is therefore about a quarter of the gross value, and the gross edge is not robust to costs for many specifications.

Figure 3. **Net annual mean return against average daily turnover under a 5 basis point cost**



Source: reviewer's calculation (generated), using gross mean returns from Table I of the paper. Assumes costs linear at 5 basis points per unit one-way turnover and 252 trading days; the holding cost is ignored and the portfolio is not retrained, so the lines are an upper bound for a fixed strategy.

Figure 3 is an illustrative break-even calculation, not a result from the paper. Because 5 basis points times 252 days equals 12.6% of gross exposure per unit of daily turnover, a gross mean of 8.7% is wiped out at an average daily L1 turnover of $8.7/12.6 \approx 0.69$, a PCA 5 mean of 14.3% at 1.13, and a Fama-French 5 mean of 4.6% at 0.37. The paper reports the turnover only as time-series plots (Figure 6), so whether the unconstrained strategy lies on the safe side of these thresholds cannot

be established from the text, but the drop from Sharpe 4.16 to about 1.1 after retraining suggests that the unconstrained strategy turned over substantially.

EVIDENCE Persistence and sparsity: with overlapping holding periods the Sharpe ratio halves after about seven trading days and is still above one at 22 days, but with non-overlapping trades it is about 0.5 at one month (Figure 12). The weights are approximately normal around zero, mostly within plus or minus 0.5% per stock (Figure 8), and 10% of stocks suffice for a Sharpe above two (Figure 9). Simple high-minus-low reversal on residuals reaches Sharpe ratios of at most about 0.3 (Figure 11).

5.6 What the network learns

PAPER The eight local filters are labelled by the authors as two up-trend, two down-trend, three reversal, and one flat pattern. The four attention heads are interpreted as early reversal, negative reversal and dampened versions. During down-trends the policy attends to the latest ten days, and during up-trends to the first twenty days of the window, which the authors link to the saying that markets take escalators up and elevators down.

CRITIQUE These interpretations come from projections of nonlinear filters onto two-dimensional linear ones, from sine-wave probes, and from average gradients on a single model fitted once on the first eight years. They show what the network responds to; they do not show that the response is economically causal or stable across refits.

6 Robustness and Additional Tests

Table 3. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Dependence on lookback length	L of 60 days instead of 30 (Tables V, VI)	Sharpe 3.5 to 3.9 for PCA and IPCA with K of 3 to 5; essentially unchanged	Partly. Two lookbacks only, and both short.
Dependence on refitting	Fixed model trained once on 4 or 8 years (Tables VII, VIII)	IPCA 5 Sharpe 2.1 (4 years) and 2.6 (8 years), alphas still significant	Partly. Shows signal is not pure refit overfit, but the drop is large.
Hyperparameter overfit	16 configurations on 1998 to 2001; five alternative networks (Tables A.III to A.V)	Validation Sharpe 3.5 to 4.2; alternatives 2.7 to 4.4 on the full sample	Partly. Alternatives are described as best performing on 2002 to 2016, so they are chosen on the test sample.
Premium disguised as arbitrage	Alpha against eight factors (Table II)	Alpha close to mean, R-squared of 1 to 4% for most K	Partly. Excludes listed factors, not liquidity risk or spread compensation.
Costs and shorting	5 bp turnover plus 1 bp short fee, retrained (Tables IX, A.X)	Sharpe 0.94 to 1.24 for IPCA; PCA negative for K of 10 and 15	Partly. Costs are low and static; no market impact or borrow constraints.

Concern	Test reported	Result	Does it resolve the concern?
Concentration and industry bets	Weight histogram, sparse portfolios, industry shares (Figures 8 to 10, A.5)	Diversified weights, industry share within 20% of population	Yes for diversification; not a test of capacity.
Weak benchmarks	Ablations OU plus FFN and raw FFN (Table A.IX)	Sharpe minus 0.3 to 1.6, below CNN plus transformer	Partly. No comparison with an LSTM, a plain MLP of equal size on FFT, or a gradient-boosted tree.
Crisis fragility	Visual cumulative returns around August 2007 and 2011 to 2012 (Figure 5)	No large drawdowns reported	No. Figure only; no stress-period statistics are tabulated.

Source: constructed by the reviewer from the tables of the paper (official numbers; assessments are the reviewer's critique).

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Within the 550-stock sample and 2002 to 2016, the CNN plus transformer policy beats the OU threshold and Fourier plus feedforward policies gross of costs, at every K and for every factor family tried, by factors of roughly two to four in Sharpe ratio. The strategy returns have almost no loading on the eight listed factors. Feeding raw returns instead of residuals sharply reduces performance (1.64 against up to 4.17).

SUPPORTED BUT CONDITIONAL

That the signal extractor, not the factor model, is the most important element. This holds for K of at least three and for the three benchmark architectures chosen. That profits survive costs: this holds for IPCA residuals under a 5 basis point cost with retraining, but not for PCA with K of 10 or 15. That the signals persist for days to weeks, which depends on overlapping rather than separate positions.

SUGGESTIVE

That the profits compensate arbitrageurs for enforcing the law of one price rather than liquidity provision. That the learned attention heads represent economically meaningful trend and reversal factors and an asymmetric response to up and down moves.

> **NOT ESTABLISHED**
>
> That a Sharpe ratio above four is attainable in practice. That arbitrage profits did not decline over time (the claim of non-declining profitability rests on plots and on a rolling-refit design). That the result generalises beyond large-cap US stocks, other periods, or other frequencies.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- A unifying decomposition that makes methods comparable, which matters because it separates the effect of factor choice from signal choice.
- Robustness to lookback, refit frequency, hyperparameters and cost model, which supports the view that the finding is not a single tuned specification.
- Interpretable components, and diversified weights across industries.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Training on the portfolio objective with a normalised book is economically coherent: it prevents the model from winning on scale, and the comparison of Sharpe versus mean-variance objectives shows the control the user has over leverage.
- The ablation structure (OU plus FFN, raw FFN, Fourier plus FFN) isolates the value of the signal block better than most machine learning finance papers.
- Reported t-statistics are internally consistent with Sharpe ratios and sample length, which indicates no obvious arithmetic error in the headline tables.
- Disclosure of failures: the PCA results with frictions turn negative and are reported rather than omitted.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The headline gross Sharpe ratio rests on daily close-to-close reversal in large caps, executed at the same close, with proportional costs of 5 basis points and no market impact. **Why it matters.** Short-horizon residual reversal is where bid-ask bounce, inventory pressure, and liquidity provision live, and these are the components least capturable by a marginal arbitrageur. **Potential bias.** Gross returns are overstated as a measure of exploitable arbitrage, and the gap between Sharpe 4.16 and about 1.1 is itself evidence of how sensitive the economics are to the cost assumption. **How it could be tested.** Re-run with signals formed at close and trades at next open or with a one-day delay, with spread-dependent costs from TAQ, and with an impact term; report the share of profit from the lowest spread tercile.

OTHER WEAKNESSES

- Multiple testing: the number of models, factor families and K values explored is large relative to a single one-year validation window. The potential bias is an inflated best Sharpe ratio, testable with a deflated Sharpe ratio (Bailey and Lopez de Prado, 2014) or a pre-registered holdout from 2017 onward.
- Test-sample selection: the alternative networks are described as the best performers on 2002 to 2016, so Table A.V is not a clean robustness test. It could be tested by selecting them on 1998 to 2001 only.
- Benchmarks: the parametric benchmark is a single parametric rule (after Avellaneda and Lee, 2010, and Yeo and Papanicolaou, 2017) with thresholds grid-searched over nine pairs; stronger benchmarks (an LSTM, a one-dimensional CNN without attention, ridge on lagged residuals) are absent, so the causal role of attention is not isolated.
- Instability: Table III shows volatilities of 93.1% and 61.2% for Fourier plus feedforward under mean-variance, which indicates unstable optimisation and calls for cautious comparison.
- Stationarity: the framework assumes that the residual process is stationary conditional on lags; the structural breaks of 2008 and 2020 are not directly tested and the 2002 to 2016 window is a single regime.
- Statistical inference: the t-statistics assume iid daily returns; autocorrelation from overlapping information and volatility clustering would reduce them, and no block bootstrap is reported for the main table.

8 Contradictions and Edge Cases

CRITIQUE When should the result disappear? The mechanism requires that residuals revert faster than costs accumulate. Equation (11) with the unit-gross normalisation implies that the strategy needs an average one-way daily L1 turnover below the break-even computed in Figure 3. Because the PCA with K of 15 case is negative after costs while it earns 7.6% gross, the gross edge in that specification must be below the turnover cost at the learned turnover, a direct internal demonstration that the result can disappear.

CRITIQUE Extreme parameter regions of the parametric benchmark show where it breaks. From Equation (6) the half-life of a deviation is $\ln 2/\hat{\kappa} = -\ln 2/\ln \hat{b}$ days. For $\hat{b} = 0.90$, $\hat{\kappa} = 0.105$ and the half-life is about 6.6 days, so within a 30-day window the process completes about four half-lives and the estimate is usable. For $\hat{b} = 0.99$ the half-life is about 69 days, longer than the window, the estimated μ is poorly identified and the stationary scale $\hat{\sigma}/\sqrt{2\hat{\kappa}}$ diverges as $\hat{\kappa} \to 0$, so the standardised score is dominated by estimation error in $\hat{\mu}$ and by the arbitrary position of the window. At $\hat{b} \geq 1$ the quantity is undefined, which the paper handles by restricting to $b < 1$. The OU benchmark's weakness is therefore a feature of small-sample AR(1) estimation, and part of the CNN gain may reflect avoiding this estimation noise rather than extracting richer patterns.

CRITIQUE Two printing inconsistencies deserve flagging. The main text writes the scaling as $\hat{\sigma}/\sqrt{2\hat{\kappa}}$ while the Appendix text writes the ratio with $\hat{\sigma}\sqrt{2\hat{\kappa}}$ in the denominator; the first is correct by Equation (6). The residual map in Equation (2) annihilates factor exposure only if $w^{F\top}\beta = I_K$, which holds for traded factors but only approximately for rolling-window estimates, and the authors acknowledge that out-of-sample residuals are not mechanically orthogonal to the factors.

CRITIQUE Multiple-testing arithmetic shows that data mining cannot alone generate the headline. If annual Sharpe ratios of null strategies were iid normal with standard deviation $1/\sqrt{15} \approx 0.26$ over 15 years, the expected best of 100 trials is roughly 2.5 standard deviations, or about 0.65. An observed 4.16 is 16 standard deviations from zero. Data mining may inflate the second decimal of the ranking but cannot generate the level; the dominant uncertainty is execution, not selection (assumptions: iid returns, 100 independent trials, generated by the reviewer).

CRITIQUE Alternative mechanisms that would produce the same pattern include bid-ask bounce and stale closing prices, end-of-day order imbalances from index funds and ETF creation, and liquidity provision to institutional order flow with inventory risk. The asymmetric response to down and up moves is also consistent with leverage-related price pressure in sell-offs and with the known larger reversal after price declines. None of these is tested.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Gatev, Goetzmann and Rouwenhorst (2006)	Distance-method pairs trading profits from convergence	Preceded; classical special case with two stocks	Gives the profitability baseline and the equilibrium-consistency argument the authors invoke.
Avellaneda and Lee (2010)	PCA residuals traded with OU thresholds	Preceded; parametric benchmark	Defines the benchmark that the paper beats by a factor of four in Sharpe.
Kelly, Pruitt and Su (2019)	Conditional latent factors with characteristic loadings	Foundation for the best factor family	Explains why IPCA residuals give the highest Sharpe ratios.
Gu, Kelly and Xiu (2020)	Machine learning for return prediction, trained on prediction loss	Complementary; contrasts prediction with trading objectives	Marks the difference between predicting risk premia and trading residual deviations. Reviewed in this library.
Chen, Pelger and Zhu (2022)	Deep learning with a no-arbitrage objective, 46 characteristics	Methodological sibling; supplies the data	Same research team and the same idea of training on an economic objective.
Vaswani et al. (2017)	Self-attention for sequences	Architecture borrowed	Source of the transformer block; reviewed in this library.
Lehmann (1990); Jegadeesh (1990)	Short-horizon reversal in returns	Preceded; alternative explanation	Reversal profits are the same object, linked to liquidity provision.

Paper	Shared idea	Relationship	Why it matters
Nagel (2012)	Reversal returns as compensation for liquidity provision	Contradicting interpretation	If correct, the paper's profits are risk compensation rather than arbitrage.
Khandani and Lo (2011)	August 2007 quant losses from crowded reversal trades	Related evidence	The authors report near immunity to the August 2007 episode; Khandani and Lo document large losses for simpler reversal strategies, so the contrast needs explaining.
Shleifer and Vishny (1997)	Limits of arbitrage with capital constraints	Theoretical background	Explains why arbitrage profits persist for days to weeks.
Bailey and Lopez de Prado (2014)	Deflated Sharpe ratio for multiple testing	Tool for the critique	Provides a way to check that the headline Sharpe survives selection.

Source: reviewer's synthesis; relationships are the reviewer's judgement, bibliographic details in the References.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** execution realism. *Why it matters:* the economic claim depends on net profits. *Existing limitation:* flat 5 basis point cost, no impact, no borrow constraints. *Way forward:* intraday data, spread-based costs, and a capacity curve.
- **Gap:** source of profits. *Why it matters:* separates arbitrage from liquidity compensation. *Existing limitation:* no decomposition by spread, volume or time of day. *Way forward:* sort profits by liquidity measures and by proxies of order imbalance.
- **Gap:** post-2016 behaviour. *Why it matters:* the claim of non-declining profits is untested for recent years. *Existing limitation:* sample ends in 2016. *Way forward:* extend to 2017 to 2026 with a frozen specification.

10.2 Literature-level gaps

- **Gap:** no agreed cost-aware benchmark for machine learning strategies. *Why it matters:* performance gaps between papers cannot be compared. *Existing limitation:* each paper uses its own cost model. *Way forward:* a common net-of-cost evaluation protocol with a published cost schedule.
- **Gap:** the link between model complexity and arbitrage capacity. *Why it matters:* limits of arbitrage theory predicts persistence tied to capital. *Existing limitation:* the persistence evidence is gross and single-manager. *Way forward:* link the signal decay to arbitrageur capital flows.

10.3 Methodological gaps

- **Gap:** formal inference for flexible trading rules. *Why it matters:* Sharpe t-statistics ignore selection and dependence. *Existing limitation:* no block bootstrap or deflation. *Way forward:* stationary bootstrap and a deflated Sharpe ratio across the full search.
- **Gap:** factors designed for tradability. *Why it matters:* residuals depend on factor rebalancing. *Existing limitation:* factor construction ignores turnover. *Way forward:* jointly optimise the factors under a turnover penalty, as the authors suggest.

11 Research Opportunities

Idea 1. Execution-realistic reversal: profits by spread and timing

RESEARCH QUESTION. How much of the CNN plus transformer profit remains when signals are formed at the close and trades execute at the next open or via VWAP, with stock-specific spread and impact costs?

MOTIVATION. The critique identifies execution as the main weakness.

HYPOTHESIS. Net Sharpe falls below 1 and most of the profit is concentrated in the lowest-liquidity tercile.

ECONOMIC MECHANISM. Reversal profits are partly compensation for liquidity provision and are paid at the quoted spread.

LITERATURE CONNECTION. Nagel (2012); Hendershott, Jones and Menkveld (2011); the target paper.

DATA. CRSP daily, TAQ millisecond quotes, S&P 500 constituents, 2003 to 2024.

VARIABLES. Residual returns, quoted and effective spreads, open and close auction prices, daily volume.

METHOD. Replicate the pipeline, add execution-lag variants, and recompute profits with spread-based costs and a square-root impact model.

IDENTIFICATION STRATEGY. Within-model comparison of timing variants and cross-sectional sorts by spread; the pre-2007 to post-2007 spread decline gives variation in costs.

EXPECTED CONTRIBUTION. A net, capacity-aware estimate of statistical arbitrage returns in liquid stocks.

MAIN RISK. Data costs and the difficulty of replicating an unreleased code base.

Idea 2. Is the attention mechanism necessary?

RESEARCH QUESTION. Do equally sized one-dimensional CNN, LSTM and ridge filters on the same residuals reach comparable Sharpe ratios, after controlling for parameter counts and tuning budgets?

MOTIVATION. The benchmarks in the paper differ in capacity and representation, so the architecture claim is not isolated.

HYPOTHESIS. A tuned CNN without attention captures at least 80% of the Sharpe ratio, so the transformer adds little.

ECONOMIC MECHANISM. The usable structure is smooth short-lag trend and reversal, which local filters can capture.

LITERATURE CONNECTION. Vaswani et al. (2017); Gu, Kelly and Xiu (2020).

DATA. CRSP daily residuals from IPCA and PCA, 1998 to 2024; Kenneth French factors.

VARIABLES. Residuals, lagged cumulative residuals, model architecture and parameter counts.

METHOD. Matched-budget model comparison with the same objective, with block bootstrap confidence intervals on Sharpe differences.

IDENTIFICATION STRATEGY. Randomised seeds and matched random search budgets; paired differences in out-of-sample returns.

EXPECTED CONTRIBUTION. Clarifies which part of the architecture earns the gain.

MAIN RISK. Results depend on tuning; the difference may sit inside the sampling noise.

Idea 3. Deflated Sharpe ratio and capacity of ML statistical arbitrage

RESEARCH QUESTION. What is the deflated net Sharpe ratio of the paper's strategy family after accounting for the full search, and how fast does it decline as assets under management grow?

MOTIVATION. Multiple testing and capacity are the two inference gaps.

HYPOTHESIS. The deflated net Sharpe ratio is positive but below 1, with capacity of a few hundred million dollars.

ECONOMIC MECHANISM. Impact and crowding reduce returns as size grows.

LITERATURE CONNECTION. Bailey and Lopez de Prado (2014); Shleifer and Vishny (1997).

DATA. CRSP, Compustat characteristics, TAQ, 13F holdings as a crowding proxy.

VARIABLES. Strategy returns, number of trials, turnover, impact parameters.

METHOD. Deflated Sharpe ratio, stationary bootstrap, capacity simulation under square-root impact.

IDENTIFICATION STRATEGY. Variation in impact estimates by stock and period; placebo with permuted residual order.

EXPECTED CONTRIBUTION. A defensible net performance range.

MAIN RISK. Impact parameters are uncertain.

Idea 4. Out-of-period test: 2017 to 2026 and the post-meme equity market

RESEARCH QUESTION. Did the profits decline after 2016, with the rise of retail flow, zero-commission trading and ETF dominance?

MOTIVATION. The paper claims non-declining profits but ends in 2016.

HYPOTHESIS. Profits fall in the later period, and fall most for stocks with high ETF ownership.

ECONOMIC MECHANISM. Faster capital and more passive flow change the speed of mean reversion.

LITERATURE CONNECTION. Khandani and Lo (2011); the target paper; limits of arbitrage.

DATA. CRSP 2017 to 2025, ETF holdings, retail order flow proxies.

VARIABLES. Residual returns, ETF ownership, short interest, volume.

METHOD. Frozen-parameter replication with annual refit; subperiod and interaction tests.

IDENTIFICATION STRATEGY. Differences across stocks in ETF ownership and the introduction of zero-commission trading in 2019.

EXPECTED CONTRIBUTION. Direct evidence on the decay of arbitrage.

MAIN RISK. Structural changes may dominate and the original code may not be available.

Idea 5. Cross-asset extension: sector ETFs and international equities

RESEARCH QUESTION. Does the residual-signal pipeline transfer to European or Japanese large caps where spreads, short sale rules and market structure differ?

MOTIVATION. Short sale and cost differences provide a natural test of the liquidity explanation.

HYPOTHESIS. Gross Sharpe is similar but net Sharpe is lower in markets with higher costs and tighter shorting.

ECONOMIC MECHANISM. Frictions in shorting and lower liquidity slow arbitrage and increase persistence.

LITERATURE CONNECTION. Gatev et al. (2006); Shleifer and Vishny (1997).

DATA. Datastream or Compustat Global for daily returns, 2002 to 2024.

VARIABLES. Residual returns, local factors, borrow fees.

METHOD. Apply the pipeline per market with local IPCA factors.

IDENTIFICATION STRATEGY. Cross-country differences in shorting bans and tick sizes, including the 2008 and 2011 shorting bans.

EXPECTED CONTRIBUTION. External validity and a test of the frictions mechanism.

MAIN RISK. Data quality and incomplete borrow cost data.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Execution realism	4	3	3	4	5	5	24
2. Is attention necessary	3	5	5	4	3	3	23
3. Deflated Sharpe and capacity	4	4	4	5	4	4	25
4. 2017 to 2026 test	3	5	5	3	4	4	24
5. International extension	3	3	3	3	3	3	18

Source: reviewer's subjective scoring.

The three strongest are Idea 3, Idea 1 and Idea 4. Idea 3 pairs the highest methodological quality with feasible data and gives the number the paper's economics need. Idea 1 targets the single most important weakness. Idea 4 is cheap and delivers a clean out-of-period test.

12 Learning Framework

12.1 Prerequisite concepts

- Factor models: arbitrage pricing theory, PCA and IPCA, and the meaning of a residual portfolio.
- Mean-reversion processes: AR(1) and Ornstein-Uhlenbeck, half-life, stationarity.
- Neural networks: convolution, ReLU, self-attention, softmax, and gradient-based training.
- Performance measurement: Sharpe ratio, time-series alpha, t-statistics, and transaction cost modelling.

12.2 Core concepts learned

- Statistical arbitrage as a three-part problem: portfolios, signal, allocation.
- Training on a trading objective differs from training a forecast.
- Residuals are better inputs to time-series models than returns, because returns are dominated by a few factors.
- Gross arbitrage profits can be large while net profits are small.

12.3 Important equations

Equation (2) is the residual map that turns a factor model into tradable relative positions. Equation (4) is the objective that trains the signal and allocation jointly. Equation (6) links an AR(1) regression to the OU parameters and the stationary scale. Equation (10) is the attention weighting that forms global patterns. Equation (11) is the cost model that links turnover to net returns.

12.4 Methods learned

Rolling-window out-of-sample factor residuals; instrumented PCA; end-to-end Sharpe ratio optimisation with stochastic gradient descent (Adam); convolutional and attention filters for short financial sequences; ablation design to isolate model blocks; overlapping versus non-overlapping holding-period analysis; interpretation through gradient importance and synthetic probes.

12.5 Questions to test understanding

COMPREHENSION

1. Why does the paper map residual weights back to stock weights with Φ_{t-1} and normalise by the L1 norm?
2. What does the sign of the OU score $(X_L - \mu)/(\sigma/\sqrt{2\kappa})$ imply for the position?
3. Why does the Fourier transform not add information, and why does it still help the feedforward network?

ANALYSIS

1. Why is the Sharpe ratio of the raw-return model so much lower than the residual model?
2. Using 5 bp $\times$ 252, what daily turnover wipes out a gross mean of 8.7%?
3. Why are overlapping holding-period Sharpe ratios higher than non-overlapping ones?

RESEARCH

1. How would you test whether the profit is liquidity compensation rather than arbitrage?
2. What evaluation would convince you that the architecture, not the tuning, is responsible for the gain?
3. How would you add market impact to the training objective while keeping training parallel?

12.6 Papers to read next

Avellaneda and Lee (2010); Kelly, Pruitt and Su (2019); Nagel (2012); Chen, Pelger and Zhu (2022); Bailey and Lopez de Prado (2014).

13 Conclusion

13.1 One-sentence thesis

Training a convolutional transformer on a Sharpe objective over factor residuals yields a gross out-of-sample Sharpe ratio above four in liquid US stocks, and signal extraction rather than factor choice explains most of the gap to simpler methods.

13.2 Three most important insights

- Residuals, not returns, are the right object for time-series learning, because they remove the few dominant common factors and stabilise the process.
- The objective matters: optimising portfolio risk-adjusted return, rather than prediction error, is what turns a pattern into a strategy.
- Gross and net performance differ by a factor of about four, so cost modelling is the main determinant of economic relevance.

13.3 Strongest evidence

The ablation ladder in Tables I and A.IX, together with the near-zero loadings on the eight factors in Table II, is the strongest evidence: the same residuals traded by different signal blocks yield Sharpe ratios of about 1, 2 and 4.

13.4 Biggest weakness

The gross profit estimate depends on same-close execution of daily reversal with a flat 5 basis point cost, which makes the headline unrealisable as stated.

13.5 Most important research gap

A net, capacity-aware and selection-adjusted estimate of the profits to machine learning statistical arbitrage.

13.6 Best research opportunity

Rerun the pipeline with next-open execution, spread-based costs and impact, and report a deflated net Sharpe ratio on a frozen specification for 2017 to 2025.

13.7 Three papers to read next

Avellaneda and Lee (2010); Nagel (2012); Chen, Pelger and Zhu (2022).

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